RESEARCH METHODS MEDIATION, MODERATION AND CONDITIONAL PROCESS MODELS

Day 4

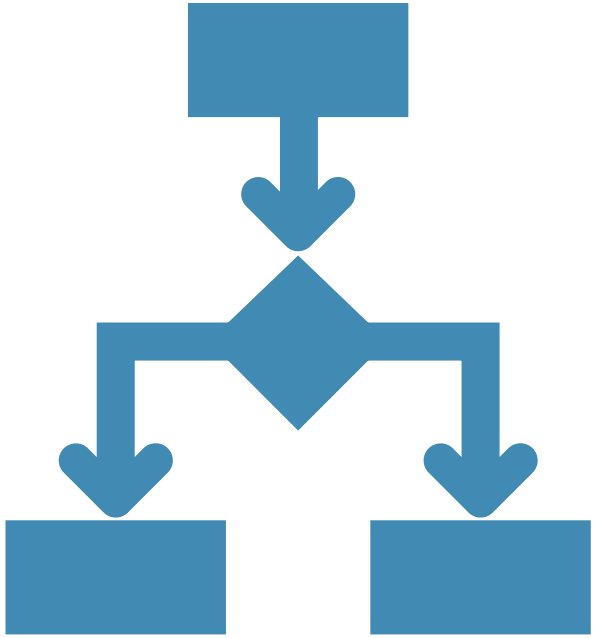
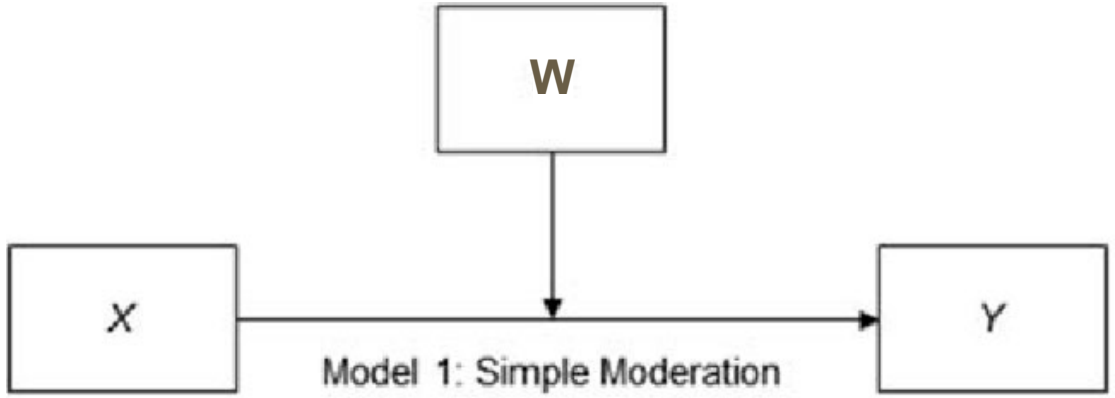


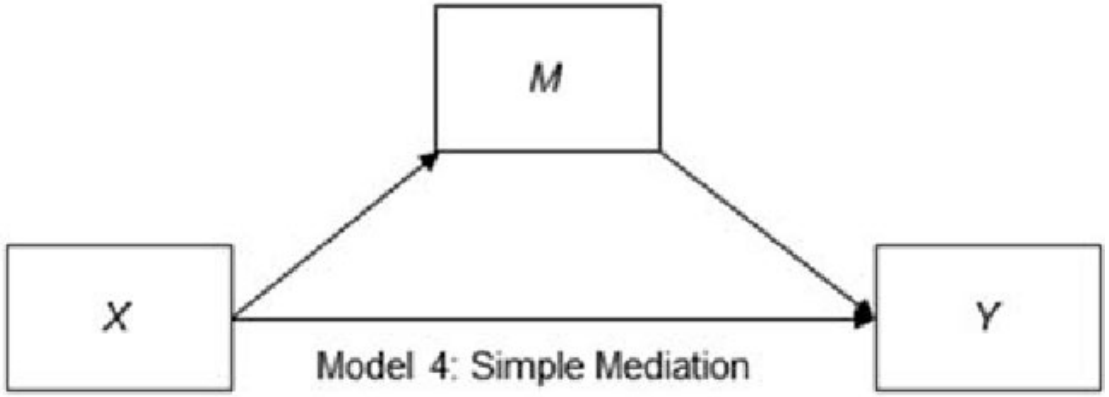
TABLE 10.5: Some Three-Variable Relationships

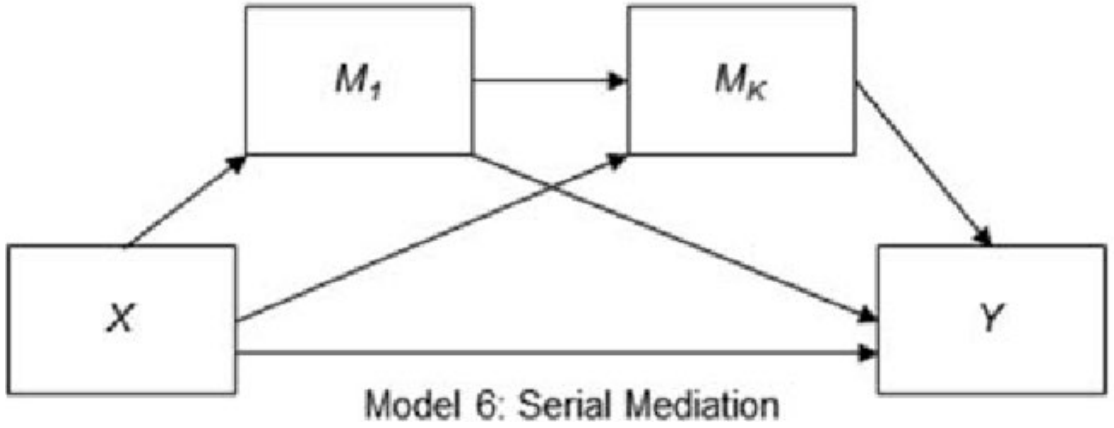
Graph	Name of Relationship	What Happens after Controlling for X_2	
$X_2 \nearrow X_1$, $X_2 \searrow Y$	Spurious $X_1 Y$ association	Association between X_1 and Y disappears.	Confounding
$X_1 \longrightarrow X_2 \longrightarrow Y$	Chain relationship; X_2 intervenes; X_1 indirectly causes Y	Association between X_1 and Y disappears.	Complete Mediation
$X_2 \downarrow$, $X_1 \longrightarrow Y$	Interaction	Association between X_1 and Y varies according to level of X_2.	Moderation
$X_2 \searrow Y$, $X_1 \nearrow Y$	Multiple causes	Association between X_1 and Y does not change.	Independence
$X_1 \longrightarrow Y$, $X_1 \searrow X_2 \nearrow Y$	Both direct and indirect effects of X_1 on Y	Association between X_1 and Y changes, but does not disappear.	Partial Mediation

Interaction Effect (Moderation)



Mediation Effect

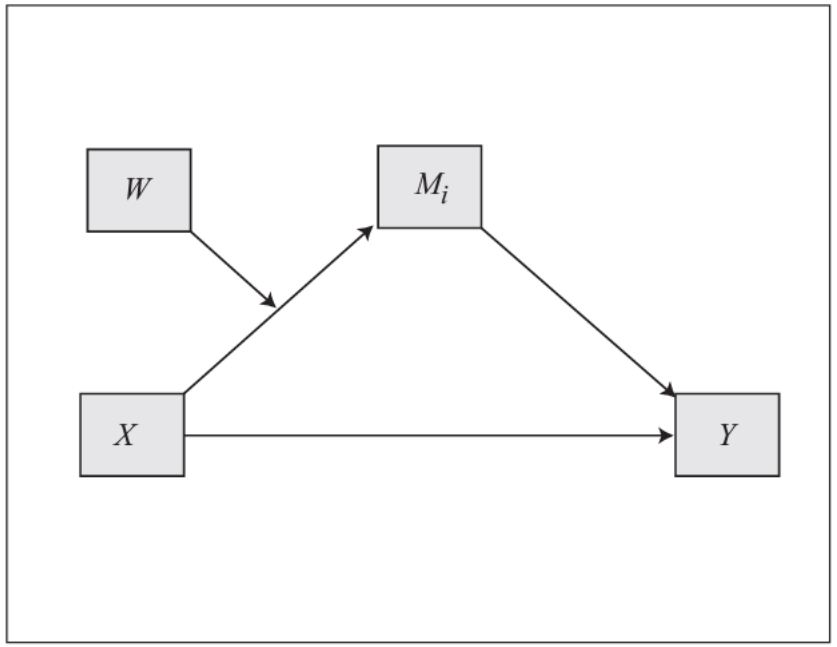




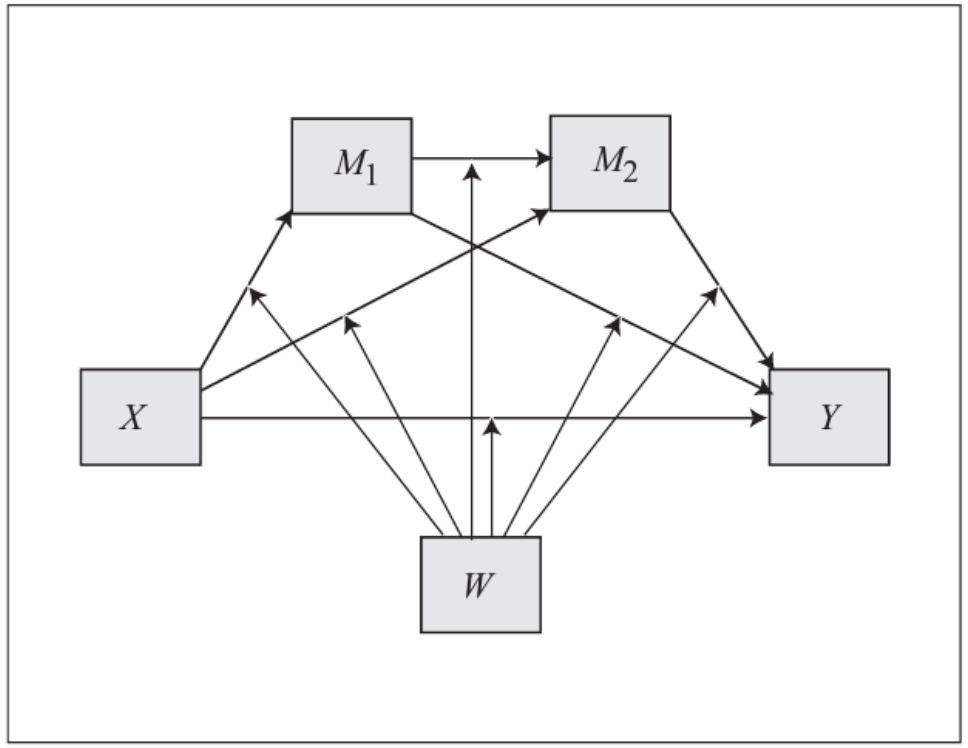
The Process Macro by Andrew Hayes

- templates.pdf (mathpsy.com)

Model 7



Model 92



Ordinary Least Squares Regression

- The SLR model (which can be extended to the multivariate case) is:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad E(Y_i) = \beta_0 + \beta_1 X_i$$

$\beta_1 > 0 \Rightarrow$ Positive Association

$\beta_1 < 0 \Rightarrow$ Negative Association

$\beta_1 = 0 \Rightarrow$ No Association

Y_i	The observed response for the *i*th trial (or row of data)
X_i	A <u>known</u> constant, the level of the predictor in the *i*th trial
β_0, β_1	Unknown regression parameters
ε_i	Error term, <u>independently</u> and normally distributed, N(0, σ^2)
i	The index for the total number of cases in your data, *i* = 1, … *N*

Interpretation of Regression Equation

$$\hat{y} = b_0 + b_1 x$$

- The magnitude of the slope b_1 tells how much Y is expected to change **per unit increase** in *X*.

- Example: $\hat{y}$ = 3 + 2x
  - b_0 = 3, $b_1 = 2$
  - If *X* increases by 1, *Y* is expected to increase by ?
  - If *X* increases by 1, then the predicted value of *Y* increases by ?

Interpreting the Intercept

- The intercept b_0 tells the predicted value $\hat{y}$ when $X = 0$.
- Note: If zero is not in or near the range of values observed for X, then the intercept has no interpretation (i.e. it represents an extrapolation).
- Example: Predicting the selling price of a house (Y) based on square footage (X)
  - It would not make sense to have house with 0 square feet, in this case the intercept would not have an interpretation.

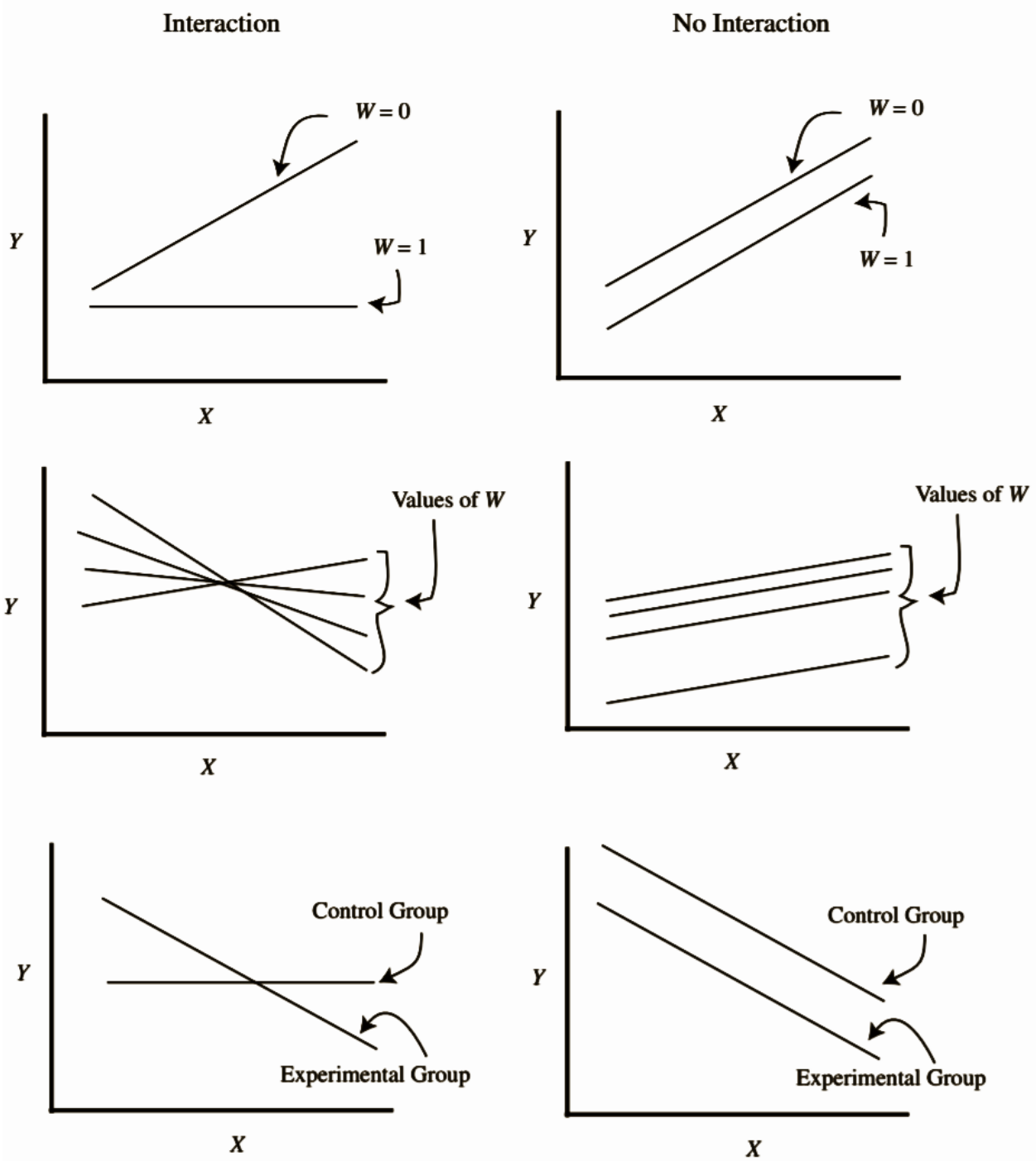
Including an Interaction Term

What does it mean for two predictors in a regression model "to interact"?

Two predictors interact if the effect on the response variable of one predictor depends on the value of the other predictor.

Review: interpreting the slope coefficient in a simple and multivariate linear regression

If an interaction effect is included in the model, the slope parameter can no longer be interpreted as the change in the mean response for each unit increase in the predictor, while the other predictors are held constant.



Two-way interactions:

Categorical x Categorical?

Continuous x Continuous?

Categorical x Continuous?

Interaction effects:

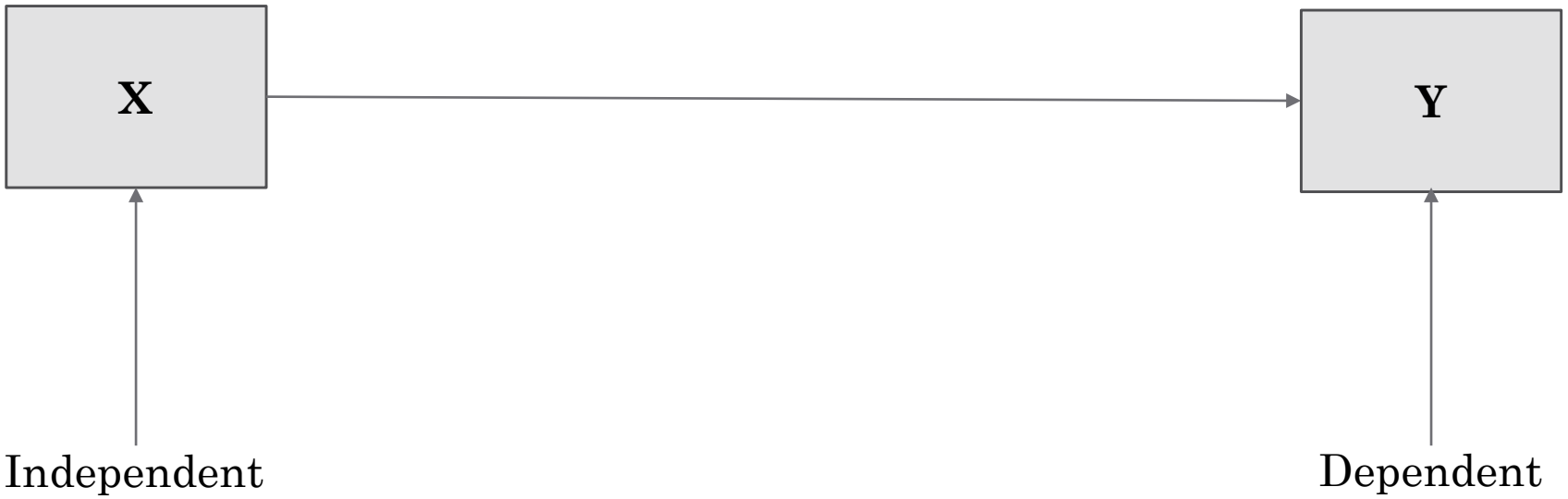
→ Change in slope (b_1)

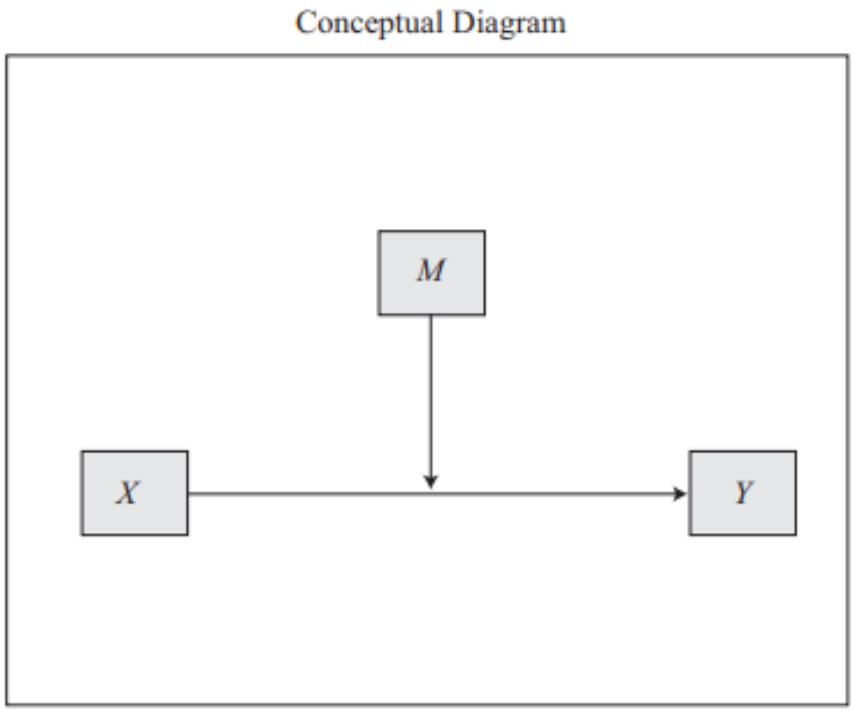
→ Change in direction (sign)

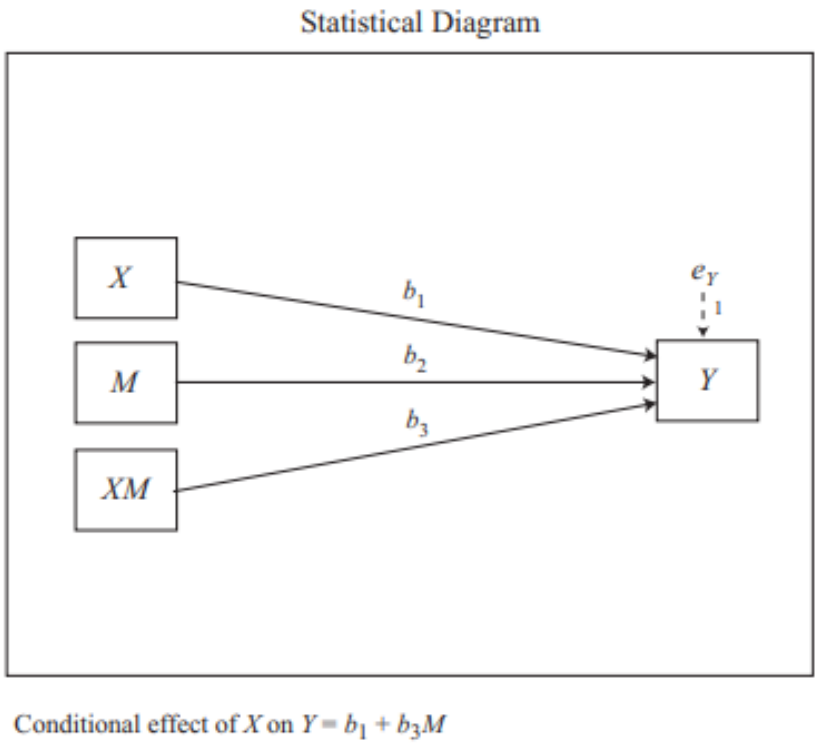
→ Change in intercept (b_0)

https://docs.google.com/presentation/d/1DojySO4VVL_GvmAmWKTg9vjE7EjIm71jpm6nW6cyN8I/edit#slide=id.p

Ways to think about relationships







Moderating Causes or "Mechanisms"

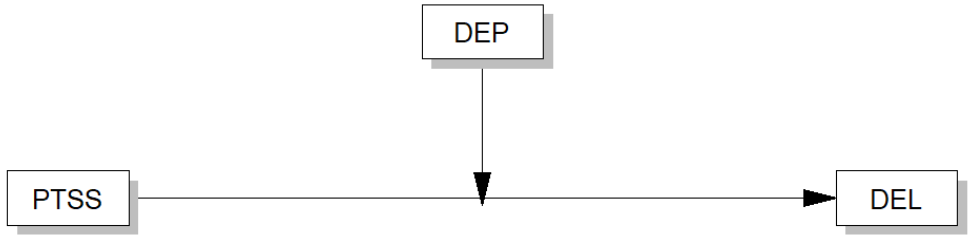
- A variable that change the nature of the relationship
- **Example**: the relationship between PTSS (*X*) and engagement in delinquent behavior (*Y*) depends on experiencing depression (*M* = *moderator*)

NSCAW I Example

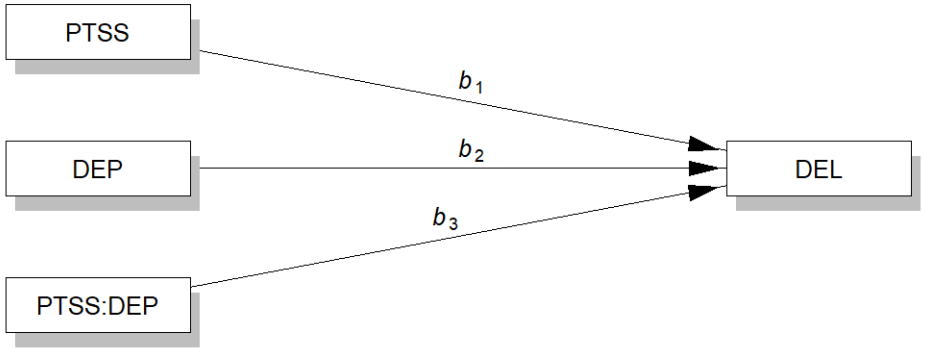
Variable	Measure
CIDI	Depressive symptoms
TRA	PTS symptoms
BCDEL	Delinquent behavior
SEXAB	Experienced sexual abuse

RQ: Do depressive symptoms moderate the effect of post-traumatic stress disorder on delinquent behavior among youth involved in the child welfare system?

Conceptual Model



Statistical Model



$$DELINQ = b_0 + b_1 PTSS + b_2 DEPRESS$$

$$DEL = b_0 + b_1 PTSS + b_2 DEPRESS + b_3 PTSS\ X DEPRESS$$

Main Effects Model

For a model without interaction, we assume that the effect of `PTSS` on `DEL` is the same regardless of the level of depressive symptoms experienced among respondents.

Interaction Model: Include a two-way interaction term of PTSS *X* DEPRESS

For a model with interaction, we want to examine if `PTSS` effects on `DEL` depend, **or are conditonal,** on level of `DEPRESS`.

Standardizing Variables

We might rescale all variables in the equation to **have a mean of 0.0 and a standard deviation of 1.0** before running a regression analysis.

Why might it be useful to standardize variables? What is the purpose of standardization?

The most common reason to use standardized coefficients is in order to have **a common scale** with which to evaluate the contribution of each of the independent variables (IVs).

- Not necessary to standardize the variables (Hayes, 2013) since standardized coefficients are hard to interpret when dealing with interaction terms.

Interaction Effects with Centering

Alternatively, we can center continuous IVs first (i.e. subtract the mean from each observation), then compute the interaction term and estimate the model. Centering is a good idea to avoid having to interpret the effects of β when *X = 0.*

- β (slope) can be interpreted as the average change in DEL when (PTSS−mean(PTSS)) increases by 1 unit, which is equivalent to the average change in DEL when PTSS increases by 1 unit. Hence, when we center the IV, the interpretation of β remains the same.
- Note that the intercept parameter has changed. Using the centered variable, the interpretation of β_0 is now the mean level of DEL for children who have PTSS scores that are equal to the sample average.

Interaction Term = Cross-Product Term

To model interaction, we can include cross-product terms of two independent variables into the multiple regression models

$$E(y) = \alpha + \beta_1 x_1 + \beta_2 x_2$$

$$\begin{aligned} E(y) &= \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 x_2) \\ &= (\alpha + \beta_2 x_2) + (\beta_1 + \beta_3 x_2) x_1 \end{aligned}$$

Interaction: $E(DEL)=\beta_0+\beta_1 PTSS+\beta_2 DEPRESS + \beta_3 PTSS^{*}DEPRESS + \epsilon$

```
Call:
lm(formula = bcdel1 ~ tra1_mean * cidi1_mean, data = moderation_example)

Residuals:
    Min      1Q  Median      3Q     Max 
-19.604  -8.252  -1.233   6.941  32.453 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)         61.104691   0.239260 255.391  < 2e-16 ***
tra1_mean            0.011538   0.022882   0.504  0.61415    
cidi1_mean           0.213284   0.022765   9.369  < 2e-16 ***
tra1_mean:cidi1_mean -0.004711   0.001568  -3.005  0.00269 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.379 on 1905 degrees of freedom
  (155 observations deleted due to missingness)
Multiple R-squared:  0.05867,	Adjusted R-squared:  0.05719 
F-statistic: 39.58 on 3 and 1905 DF,  p-value: < 2.2e-16
```

Interpret the coefficient on PTSS (TRA) and Depressive symptoms (CIDI)

What is the expected (i.e., predicted) DEL score for welfare involved youth who have a PTSS score of 45 and a Depressive symptoms score of 50?

How to interpret the interaction between PTSS and DEP?

Compare Hayes book with R result

		Consequent			
		bcdel1			
Antecedent		Coef	SE	t	p
tra1_mean	*ac'*	0.012	0.023	0.504	.614
cidi1_mean	*ac'*	0.213	0.023	9.369	<.001
tra1_mean:cidi1_mean(tra1_mean:cidi1_mean)	*ac'*	-0.005	0.002	-3.005	.003
Constant	$ibc'del_1$	61.105	0.239	255.391	<.001
Observations		1909			
R2		0.059			
Adjusted R2		0.057			
Residual SE		9.379 (df = 1905)			
F statistic		F(3,1905) = 39.580, p < .001			

Interpretation of the Intercept

The intercept tells us the expected DEL score (~61.1) for the "average" welfare-involved child:

- A child with an average level of PTSS symptoms,
- A child with an average level of depressive symptoms

Interpretation of Regression Coefficients (Main Effects)

The coefficient for PTSS (0.0115) is the average difference between the delinquency scores for children who have an additional symptom. Hence, a one unit (i.e., one additional symptom) increase in post-traumatic stress is associated with a 0.0115-unit increase in the average DEL score, holding DEPRESS constant at its mean

Interpretation of the Interaction Term

Controlling other variables constant, the results show that while a one-unit increase in PTSS is associated with a 0.0115 unit increase in the average DEL score (i.e., the main effect), the interaction term (PTSS X DEPRESS) is statistically significant at the 0.05 level, meaning that there is an interaction effect between PTSS and DEPRESS, i.e., the effect of PTSS on DEL is conditional on level of depressive symptoms the child is experiencing – the effect of PTSS on DEL is different for different values of DEPRESS.

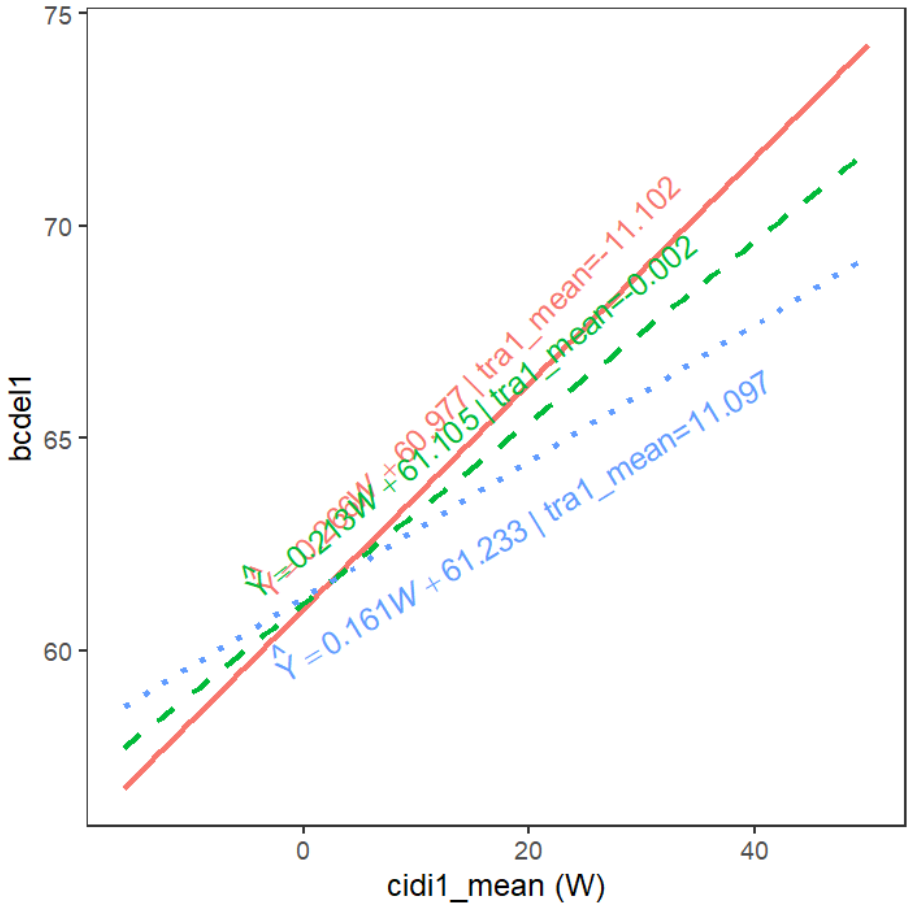
This means that, **as DEPRESS increases**, a one-unit increase in PTSS *decreases* DEL scores, holding other variables constant. This is because the coefficient on the interaction term is negative.

Alternatively:

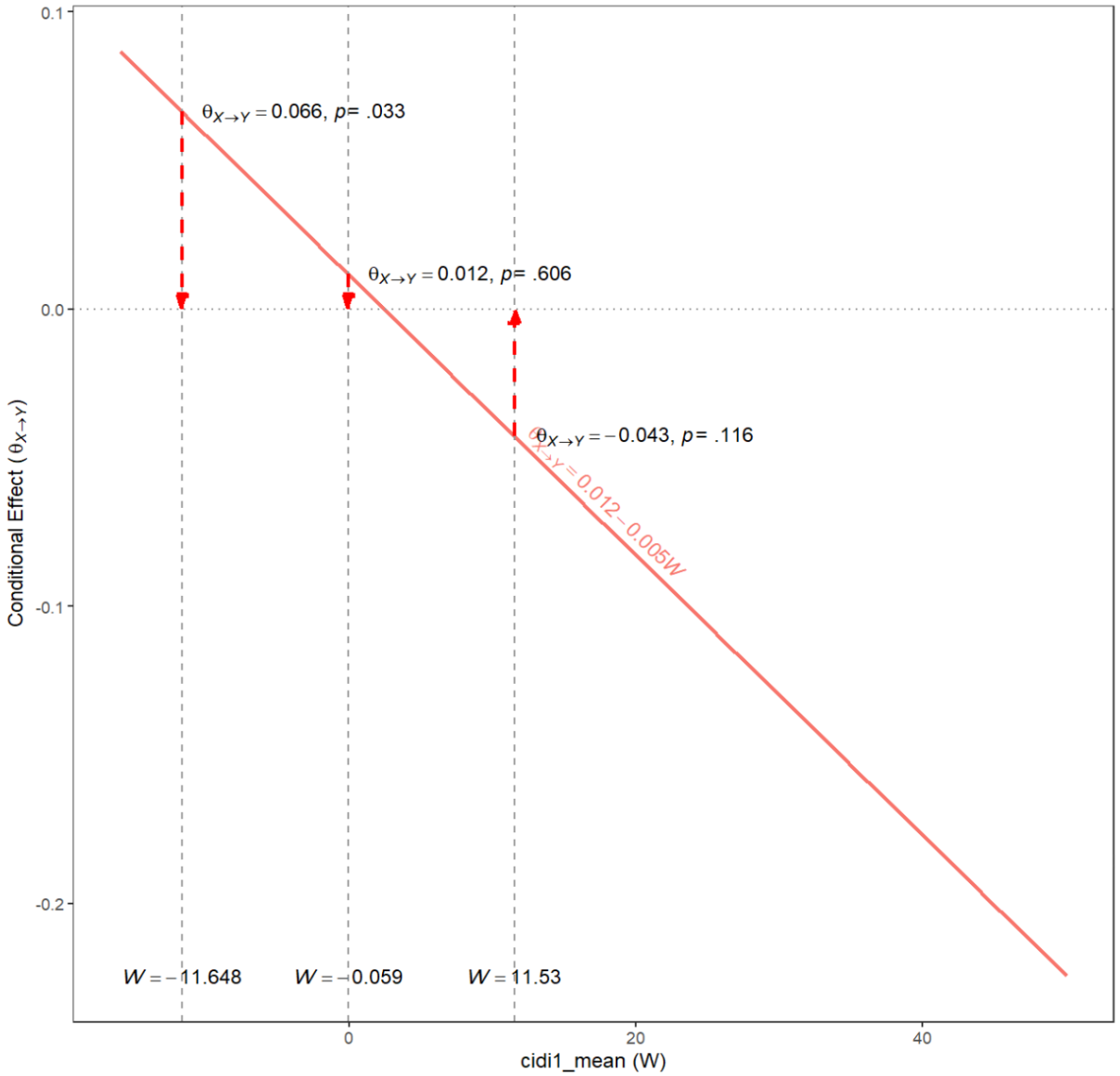
The effect of PTSS on DEL is different for different values of DEPRESS.

When PTSS increases by 1 unit, the coefficient (=slope) of DEPRESS for DEL decreases by **0.2085 units** (.2132 - .0047) = 0.2085), compared to the coefficients of DEPRESS (i.e., .2132) for children who have average level of PTSS.

```
Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)         61.104691   0.239260 255.391  < 2e-16 ***
tra1_mean            0.011538   0.022882   0.504  0.61415
cidi1_mean           0.213284   0.022765   9.369  < 2e-16 ***
tra1_mean:cidi1_mean -0.004711   0.001568  -3.005  0.00269 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

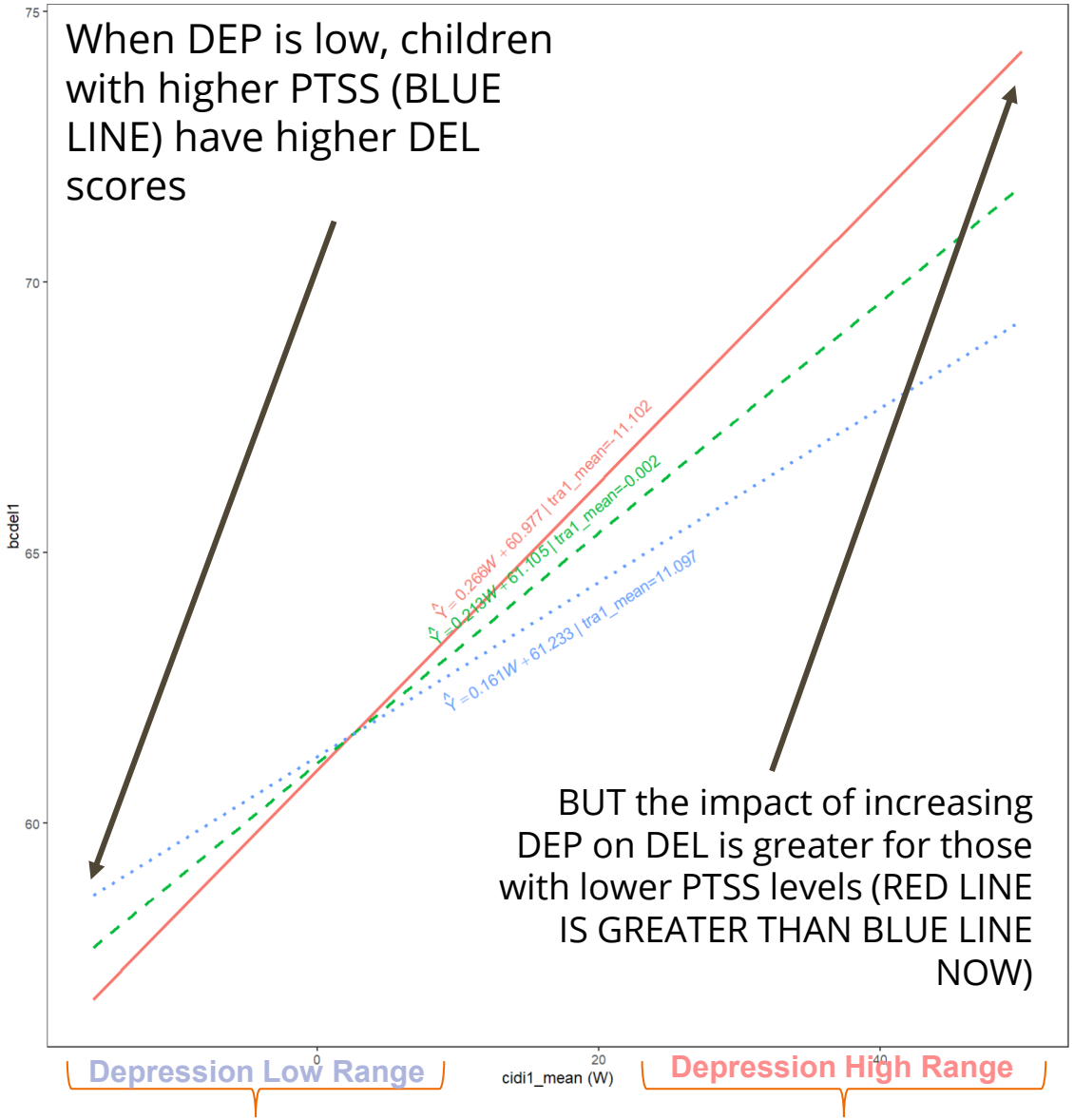


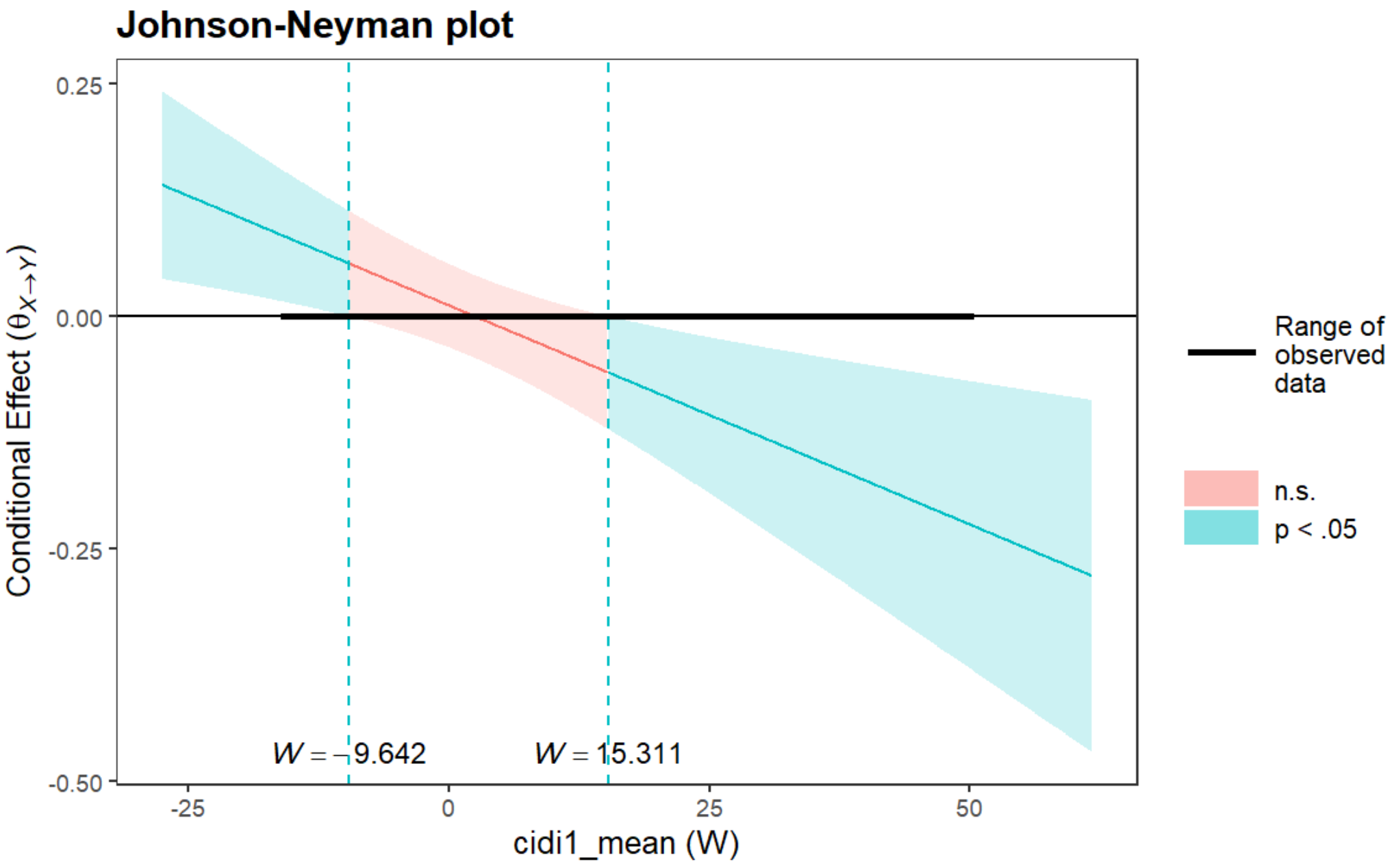
This is an odd finding: depressive symptoms are protective for delinquent behavior among youth with PTSS (?)



As depressive symptoms increase the impact of PTSS on DEL is reduced

When DEP is high, children with higher PTSS (BLUE LINE) have lower DEL scores




Johnson-Neyman plot
0.25
0.00
-0.25
-0.50
Conditional Effect ($\theta_{X \to Y}$)
$W = -9.642$
$W = 15.311$
-25
0
25
50
cidi1_mean (W)
Range of observed data
n.s.
p < .05

What is Mediation? Another way to think about causality

- Ordinary Least Squares regression examines whether independent variables have a “direct” effect on a dependent variable. Direct effects are common in public health, social work, etc. research, but “indirect” effects are also common.
- The terms “indirect effects” and “mediation” are used interchangeably. To remember what mediation is, think about how *mediators* are people who work *between* two people to resolve a dispute. In statistics, mediators come between two variables.
- Mediation is present in statistics when the effects of an independent variable on a dependent variable occur fully or partially *through* its effects on another variable.

What is Mediation?

Mediation refers to the effect of an independent variable (*X*) on a dependent variable (*Y*) is transmitted through a third intervening/mediating variable (*M*). In other words, an indirect effect is when *X* affects *M*, which in turn affects *Y*, other things being equal. Hence, we say *M* mediates the effect of *X* on *Y*.

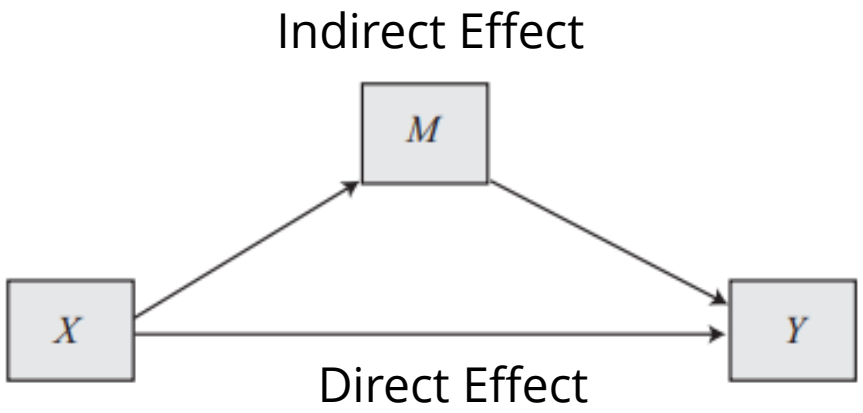
Note: *M* is just another independent variable in your model.

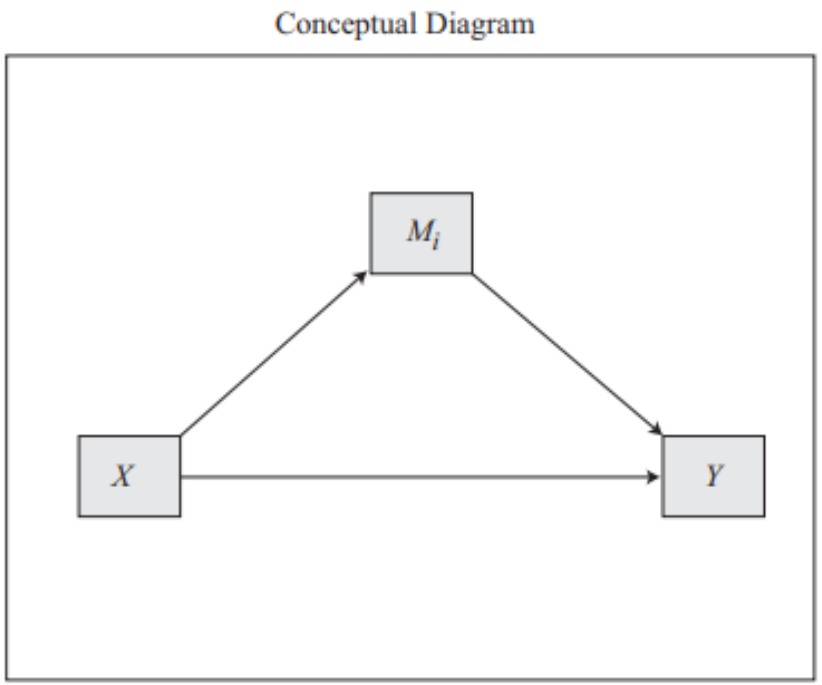
Any examples you can think of?

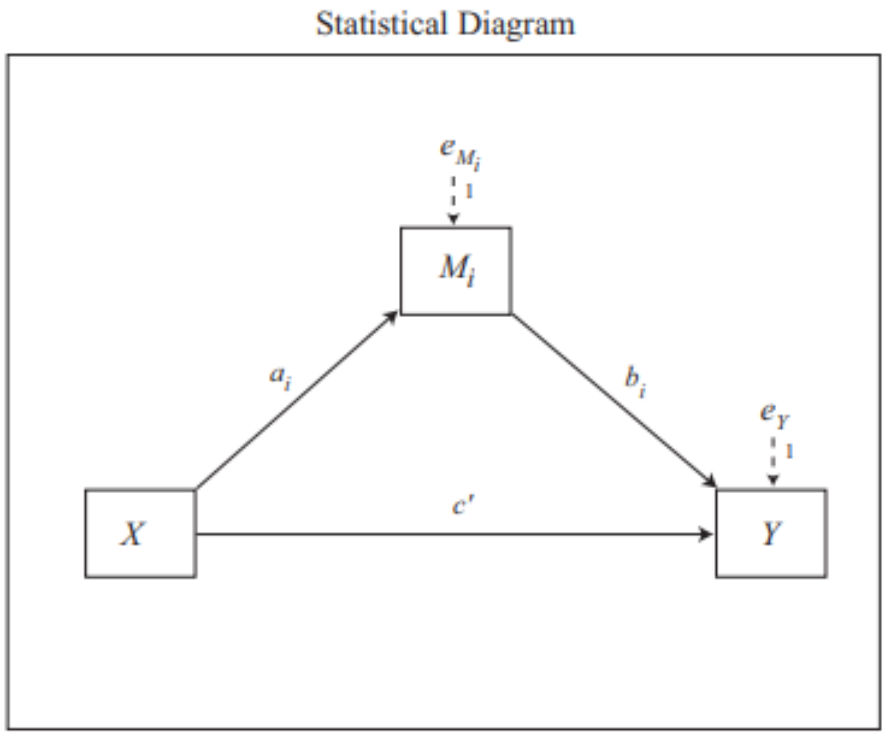
- Parent education → _____ → Poverty

What is Mediation?

- Mediation analysis is a statistical method used to evaluate evidence from studies designed to test hypotheses about how some causal *antecedent* variable *X* transmits its effect on a *consequent* variable *Y*
    - Note: Hayes is a psychologist, and they have different terminology, don't be confused, antecedent means independent variable
    - *An antecedent variable is an independent variable; consequent is dependent*
- RQ: What is the mechanism, be it emotional, cognitive, biological, or otherwise, by which X influences Y?







Mediating Causes or "Mechanisms"

- Explains the **mechanism** by which two variables are related
- **Example**: the relationship between experiencing ACEs (*X*) and delinquent behavior (*Y*) exerts itself through levels of PTSS (*M* = *mediator*)

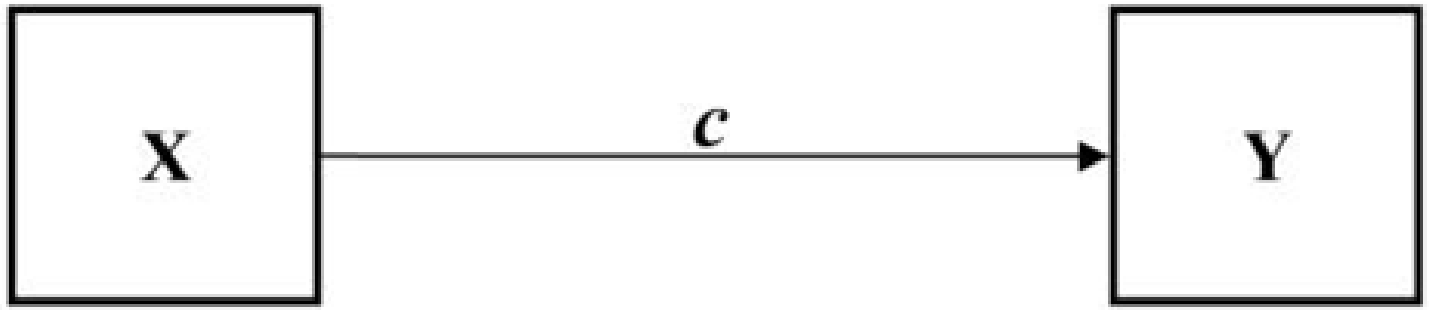
Conceptualizing a Mediation Process

- Mediation is ultimately a causal explanation hence the *assumptions are that*
  - The relationships in the system are causal
  - M is causally located between X and Y
  - X causes M, which in turn causes Y → M is located causally between X and Y
- Note about cross-sectional data

Path diagrams

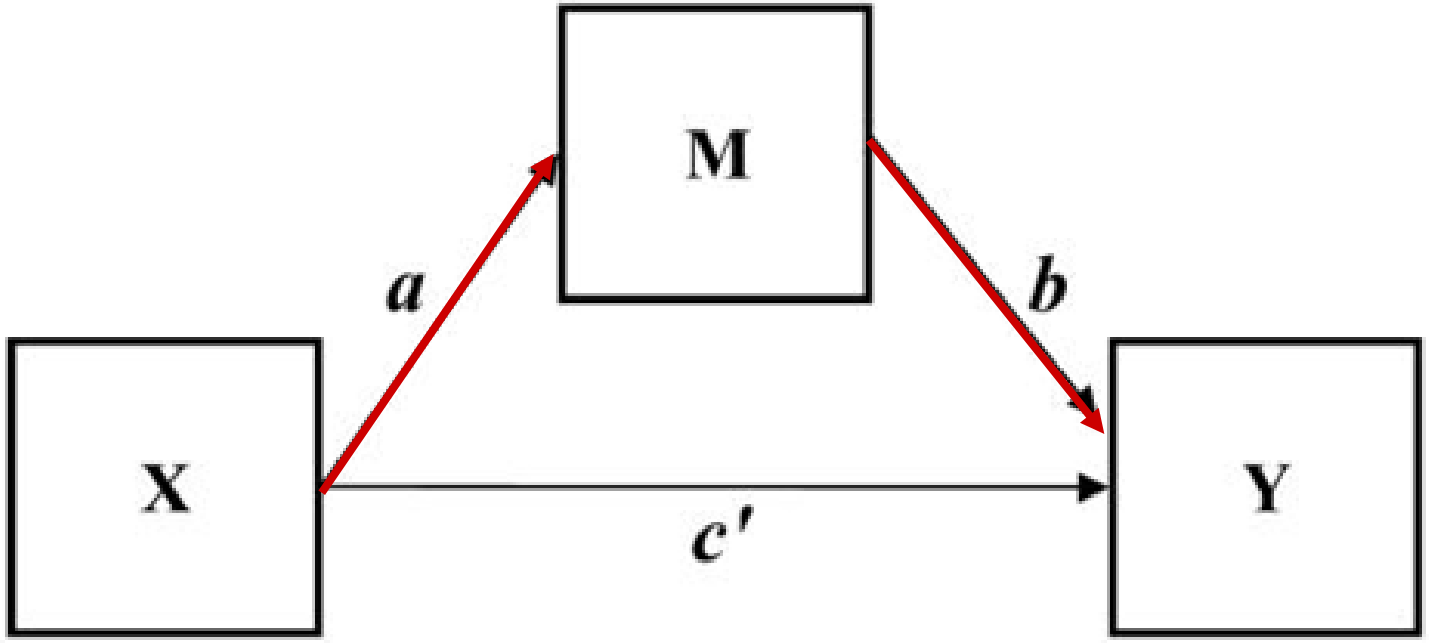
Total Effect Model:

i.



Basic Mediation Model:

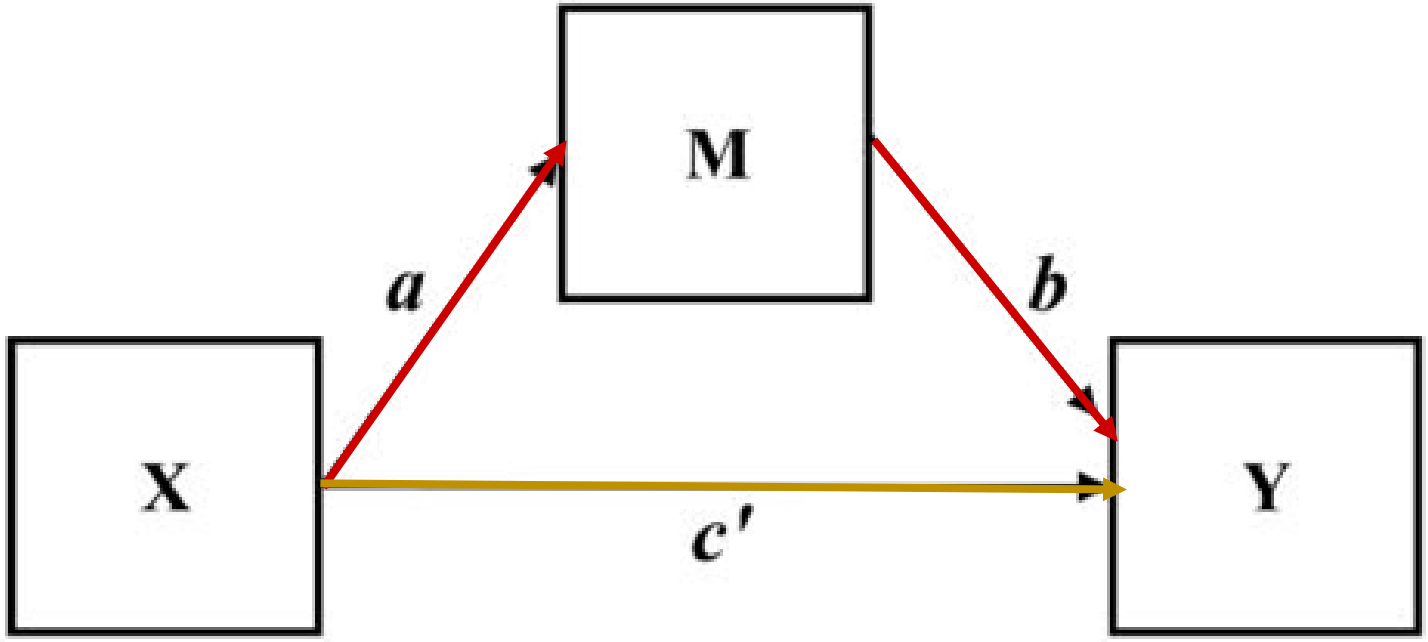
ii.



Total Effect of X on Y in Mediation Analysis

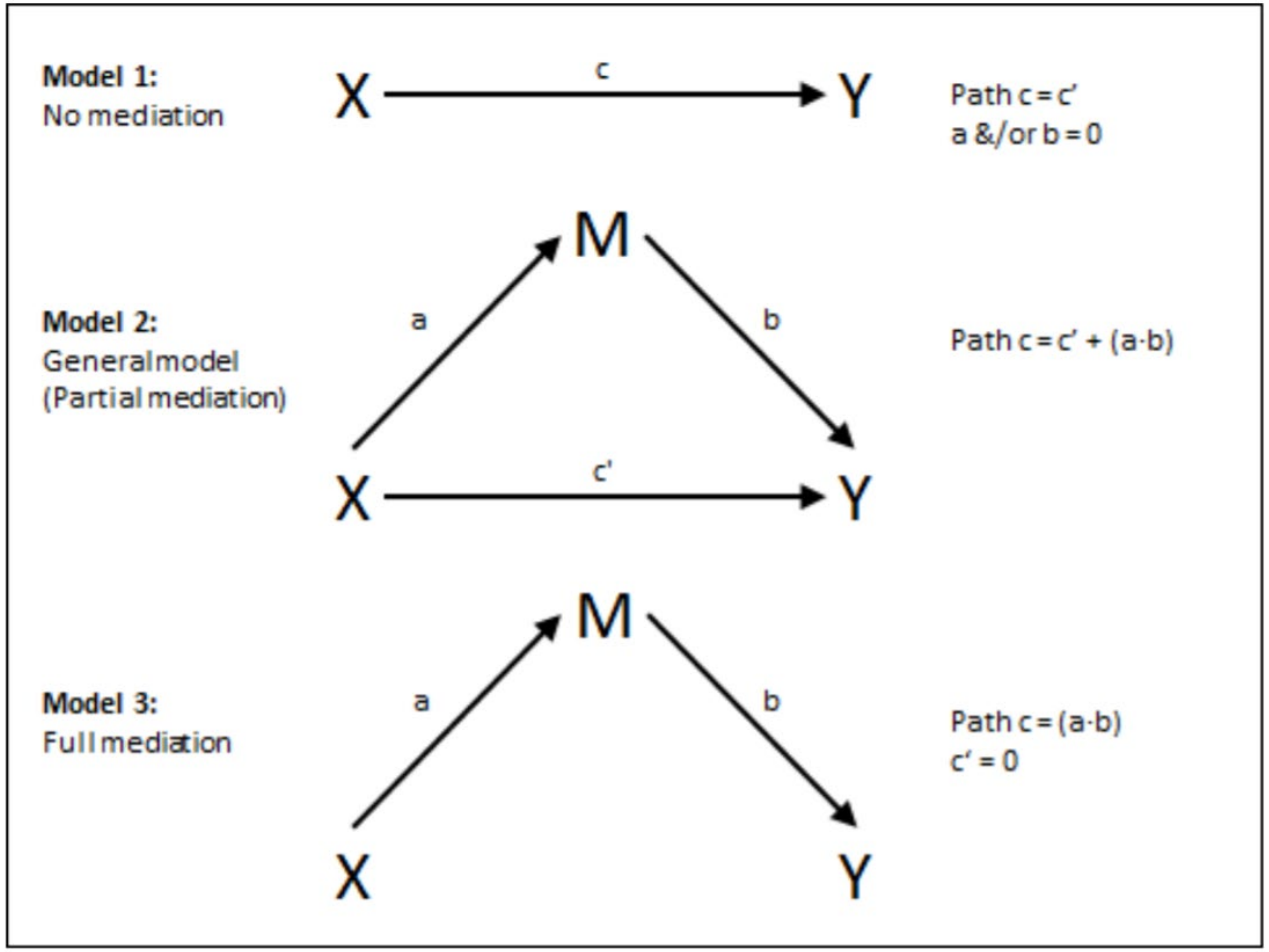
$$c = ab + c'$$

ii.



1. *c* = the total effect of *X* on *Y*
2. *c*'= the direct effect of *X* on *Y* after controlling for M
3. *ab* = indirect/mediation effect of *X* on *Y* via *M*

No Mediation, Partial Mediation & Full Mediation

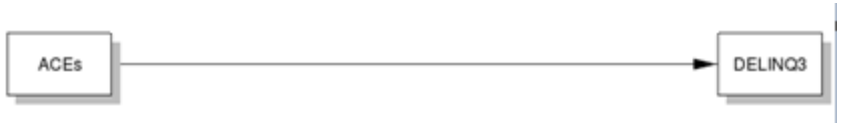


- **Model 1 (No Mediation)** shows the total effect between X and Y.
- **Model 2 (Partial Mediation)** has both direct and indirect effects between X and Y.
  (1) X relates to Y **directly** via pathway c';
  (2) X correlates with M (mediator) through pathway a; and M relates to Y via pathway b. Pathways a and b multiply to signify the **indirect** effect.
- **Model 3 (Full Mediation)** depicts the indirect effect **without the direct effect of X on Y**.

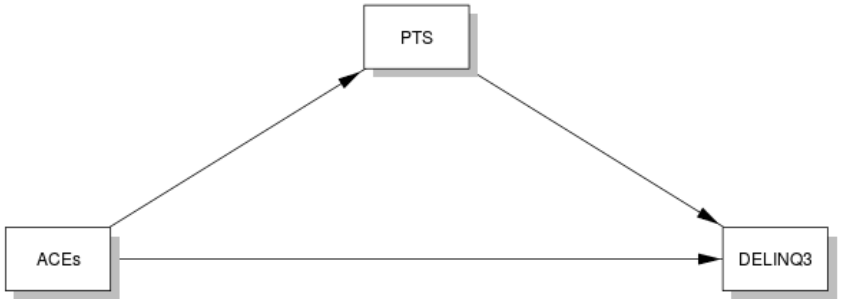
→ Check the changes in coefficients to see if mediation happens, if so, which type.

https://journals.uair.arizona.edu/index.php/jmmss/article/viewFile/17761/17484

Think about the nature of the relationships

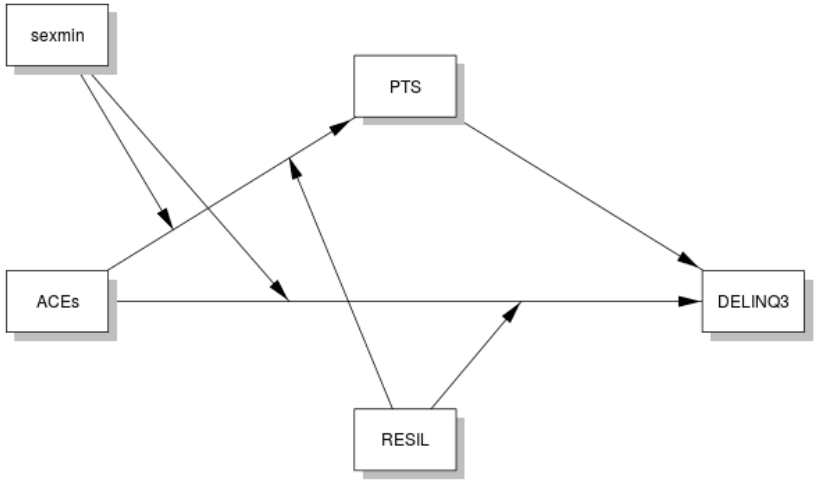


Adverse child experiences is associated with delinquent behaviors.
What is the nature of the relationship? Why might the relationship exist?



By what mechanism do ACEs increase engagement in delinquent behavior?

Be careful to understand what this means (Keep calm and read Hayes)… More ACEs results in higher levels of PTSS which in turn results in more delinquent behaviors.



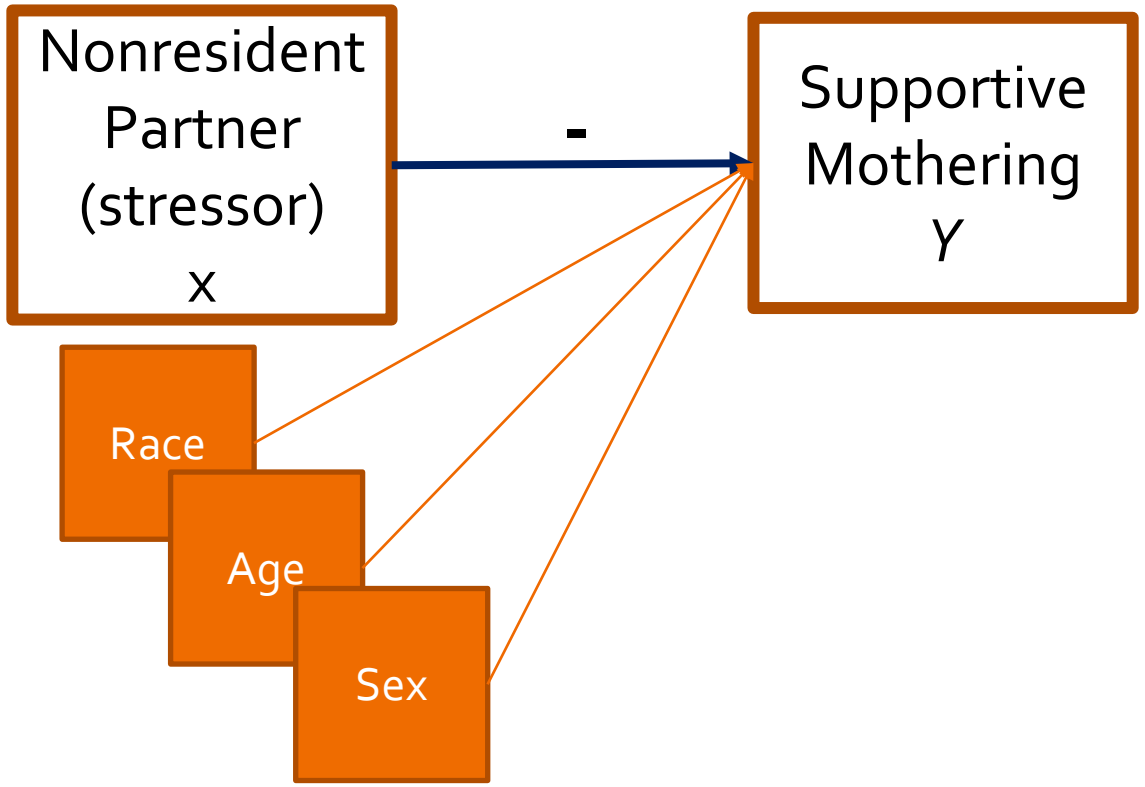
What variables might change the nature of the relation between ACEs and delinquency?

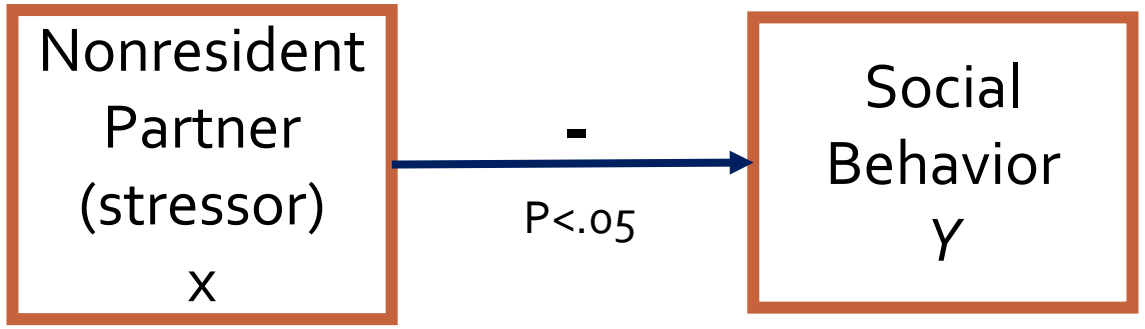
Be careful to understand what this means (Keep calm and read Hayes)… Sexual minorities with more ACEs have higher levels of PTSS which will result in more engagement in delinquent behaviors.

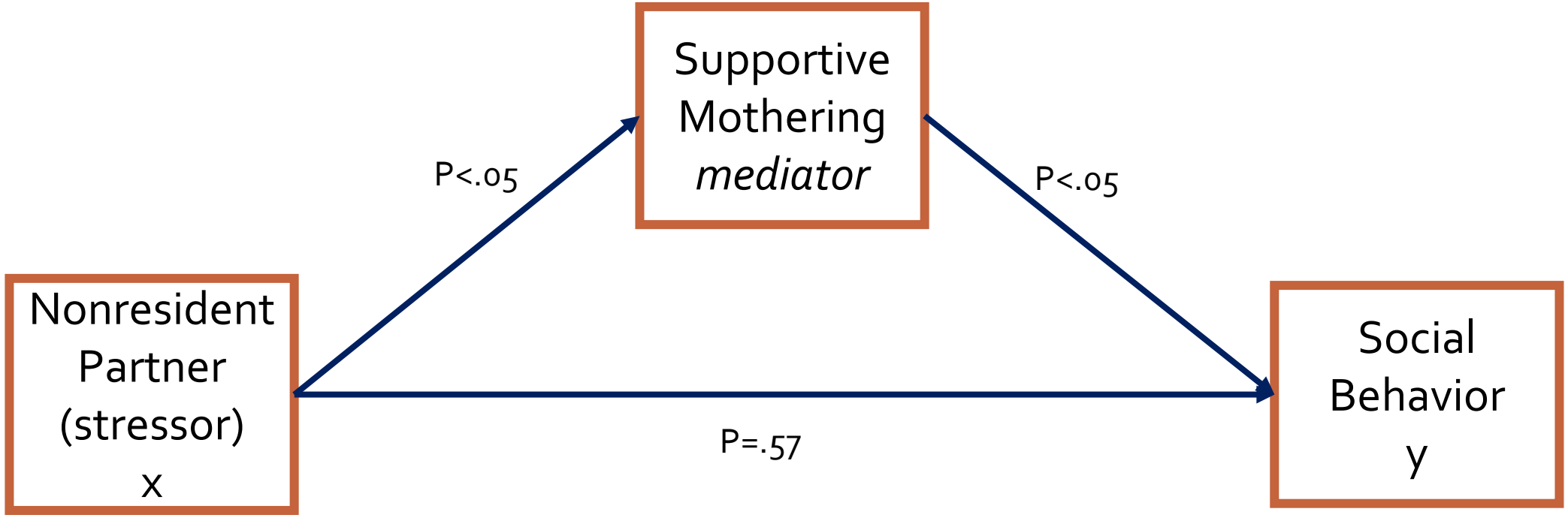
I used these methods in the following paper

- Barboza-Salerno, G. E., & Remillard, A. (2023). Early Child Adversity and Delinquent Behavior in Foster Care Youth: Do Future Expectations and Sexual Identity Moderate the Mediating Role of Posttraumatic Stress?. *Journal of Child & Adolescent Trauma*, 1-13.

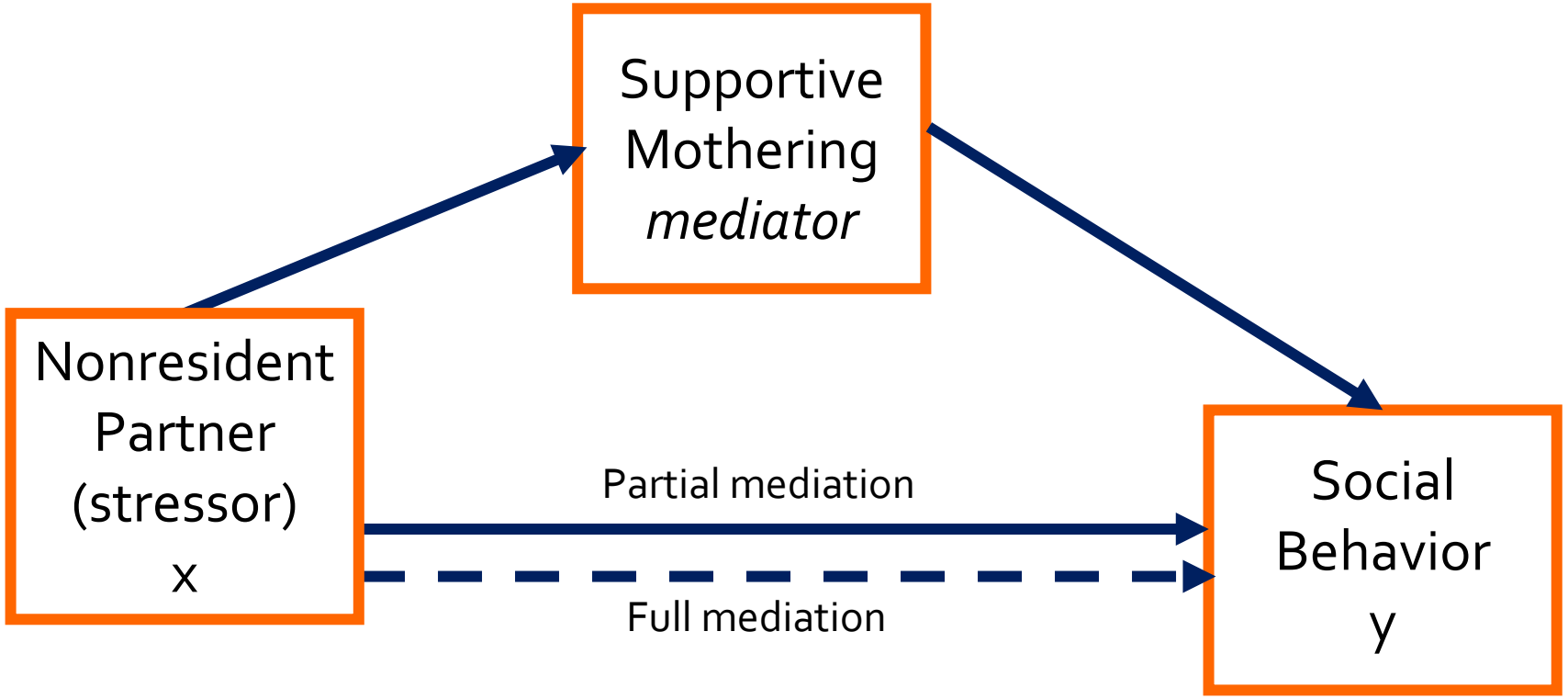
Simple Linear Regression (in a diagram)




Nonresident Partner (stressor) x
-
P<.05
Social Behavior Y


Supportive Mothering *mediator*
P<.05
P<.05
Nonresident Partner (stressor) x
P=.57
Social Behavior y

Mediation Model Based on Stress Theory



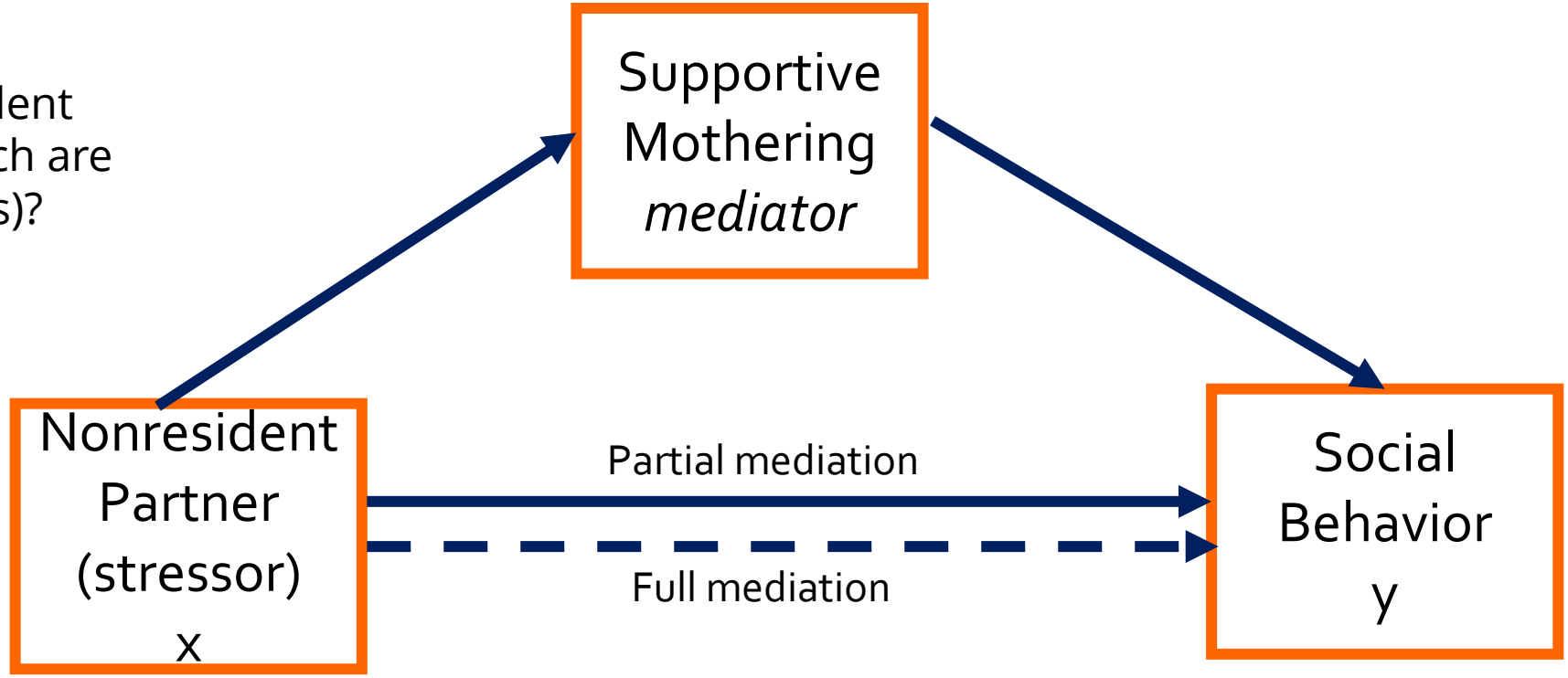
The dotted line indicates no significant direct effect.

Identifying Mediation Through Regression Results

Regression equations have only one dependent variable. We can't test a mediation model in one analysis because two variables are dependent in the figure.

Which variables are dependent variables in the model (which are predicted by other variables)?

Which variable is both a predictor and a predicted variable in the figure?



What about Suppression?

With suppression, the addition of a third variable *increases* the effect of the original independent variable on the dependent variable. Suppression is not examined as often as mediation effects, so it isn't clear how common it is.

Suppression can also be tested efficiently with "hierarchical" regression.

Evidence of suppression would be that a small or nonsignificant effect of an IV (like nonresident partner) in one model becomes larger and statistically significant with the addition of a variable (like supportive mothering) in a subsequent model.